

Minji Chloe Kim

Data Engineer / MLOps Engineer — from manufacturing analytics to production ML pipelines

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Summary

IoT engineer transitioning into data engineering — from U.S. Samsung-device manufacturing analytics to owning production ML pipelines. Built a Kafka → LightGBM → SHAP real-time inference pipeline on an 11,939-wafer SK hynix semiconductor dataset, and a hybrid-retrieval RAG agent reaching 93% in-scope accuracy over 121 live consultations. Seeking a graduate program in Data Engineering / MLOps.

Education

SoonChunHyang University

Bachelor of Science in Internet of Things. GPA [3.62/4.5]

Asan, South Korea

Mar 2021 – Aug 2026

ASAC × SK Planet T Academy — Data Engineering Bootcamp

Seoul, South Korea

Data Engineering Cohort 2

Feb 2026 – Aug 2026

- Completed an intensive data-engineering curriculum — real-time streaming (Kafka), workflow orchestration (Airflow), containerization (Docker), and model lifecycle management (MLflow) — applied directly in the SK hynix and RAG platform projects below

Experience

HDI (Asurion device-repair partner — Samsung smartphone refurbishment)

Richardson, TX

Assistant Data Analyst (J-1 Internship, full-time)

Jul 2024 – Jul 2025

- Owned recurring production-loss and station-bottleneck reporting across a 20+ station Samsung smartphone refurbishment line — hands-on with data from every station, Receiving → CellTech → Quality Verification → Shipping — building the SQL Server and Excel reports distributed enterprise-wide via a Java-based internal system for parts pre-ordering decisions
- Led a root-cause analysis of OCTA (display-assembly) scrap that lifted HDI's reclaim rate from 1.59% to 23% — past the 9.05% overall OCTA benchmark — by isolating the main defect drivers and the corrective actions to fix them
- Built a daily SIM-card usage-and-defect tracking dataset to add precision to SIM-card ordering — one of the most lead-time-sensitive parts — reducing stockout and over-order risk
- Instrumented part-room order alerts with priority scoring and end-to-end pickup-time measurement, and rebuilt the work-volume tracking dashboard, giving the floor real-time throughput and order-priority visibility

Projects

Active Manufacturing AI Platform (Team AITCH)

SK hynix (participating company) × ASAC × SK Planet T Academy

MLOps Engineer — Prediction & Retraining Pipeline (Team A of 4)

Jun 2026 – Aug 2026 (Ongoing)

- Engineering the real-time inference pipeline for Etch-process defect prediction as sole owner of Pipeline A — Kafka-based FDC trace ingestion, Step×Window feature engineering, LightGBM ensemble modeling, and SHAP Top-3 attribution — across an 11,939-wafer, 123K-row semiconductor dataset
- Diagnosed abnormally high R^2 as train/test leakage from random cross-validation, redesigned the evaluation protocol around time-based regime holdout, now adopted as the team's standard performance benchmark
- Devising Model R2R, an automated residual-bias correction system that consumes delayed ground-truth labels and updates predictions within a clamped bound, with an escalation path for out-of-bound drift
- Administering the continuous training (CT) loop — sliding-window retraining triggered on equipment maintenance cycles, champion-challenger promotion gated on regime-holdout metrics, and MLflow-based model version control

Youth Life Legal Consultation AI (RAG-based Multi-Agent System)

ASAC × SK Planet T Academy

MLOps Engineer — Supervisor Routing & Labor-Domain RAG Agent (Team of 4)

Jun 2026

- Built the Supervisor routing layer in LangGraph, auto-classifying incoming legal questions into 4 domains (labor, housing, consumer, finance) and directing out-of-scope queries to an honest-decline path
- Owned the labor-domain RAG agent (166 indexed statute articles from Korean Labor Standards Act & Minimum Wage Act), implementing hybrid BM25 + embedding retrieval to handle colloquial-to-legal term expansion
- Contributed to a hallucination-prevention harness — grounding verification via a Verifier node, hardcoded contact-data lookup, and forced disclaimer injection — achieving a 77% verification pass rate and 93% in-scope response accuracy across 121 live consultations
- Collaborated across a 4-person team under a shared CLAUDE.md convention and 3-tier Git branch strategy (feat → dev → main), with per-domain file ownership and PR-gated changes to shared modules

Chicky — Emotion-Aware AI Companion Doll (Capstone Design)

SoonChunHyang University

Project Leader — ML Pipeline & Embedded Systems (Team of 4)

Sep 2023 – Jun 2024

- Led a 4-member team building an AI companion doll that classifies children's emotional states from real-time dialogue and recommends emotion-matched audiobooks, owning both the ML pipeline and embedded communication layer end-to-end
- Trained and benchmarked 3 Korean sentence-classification models (KoBERT, KLUE-RoBERTa, KcELECTRA) on a 19,374-row, 7-class emotion dataset; diagnosed 16.7% duplicate-utterance leakage and fixed it with a group-aware dedup split, selecting KLUE-RoBERTa as final model at 93.54% macro-F1
- Deployed the classifier behind a Flask inference service consumed by the Spring Boot backend, mapping detected emotion to book-theme metadata for downstream audiobook recommendation
- Designed a dual-mode (local-socket / remote-polling) communication architecture on Raspberry Pi with automatic mDNS device discovery, integrating Azure Speech-to-Text, GPT-3.5, and camera streaming over WebSocket/TCP/HTTP

Leadership & Activities

University Student Council

Asan, South Korea

Director of Accounting

Mar 2021 – Dec 2023

- Managed departmental budgets and administrative workflows for 3 consecutive years, overseeing the transition from manual record-keeping to digital tracking systems
- Supervised a team of staff members to coordinate large-scale university events, ensuring operational efficiency and fiscal transparency
- Developed and deployed an OCR-based automation pipeline that converts bank statement screenshots into structured ledger entries, cutting ledger preparation from 3 to 1.5 days and audit verification from 5 to 2 days

Freshman Mentoring Program

Asan, South Korea

Lead Mentor

Mar 2022 – Dec 2023

- Guided 50 incoming students through academic and social transitions over 2 years, conducting regular one-on-one consultations to support student retention and satisfaction

Bucheon Braille Library

Bucheon, South Korea

Volunteer Data Typist

Jan 2023 – Present

- Transcribed printed literature into digital braille formats, ensuring 100% data accuracy to facilitate information accessibility for the visually impaired
- Coordinated large-scale text conversion projects, maintaining consistent formatting and linguistic integrity across diverse academic and literary materials

Awards & Certifications

Skills & Interests

Technical: Python, SQL, Java, R, C/C++

MLOps & ML: LightGBM, SHAP, MLflow, Apache Kafka, Airflow, Docker, LangGraph, PyTorch, HuggingFace Transformers

Tools & Platforms: Microsoft SQL Server, AWS, Tableau, Advanced Excel, Raspberry Pi, Azure Speech-to-Text, GPT API

Language: Korean (Native), English (Professional Working Proficiency)

Interests: Swimming, Violin